The Islamia University of Bahawalpur

Final Year Project Proposal

Program: BS Information Technology (BSIT)
Nature of Project: Development
Project Title: Employee Onboarding and Offboarding Workflow Tracker using Flask and SQLite
Short Title: Employee Workflow Tracker
Student Name: Syeda Hijab Fatima
Roll Number: 1086
Project Supervisor: Dr Rizwan Ali Shah

1. Project Title + Short Title Brief

Project Title: Employee Onboarding and Offboarding Workflow Tracker using Flask and SQLite
Short Title: Employee Workflow Tracker
This project is a web-based application designed to manage employee joining and leaving procedures in an organized way. It allows HR staff to create checklists, assign tasks to departments, monitor progress, and track completed and overdue tasks through a centralized system.

2. Project Brief

Many organizations manage employee joining and leaving through emails, paper checklists and spreadsheets. Tasks such as account setup, document collection, equipment handover and clearance are often missed or delayed, and HR has no clear view of progress. This project is a web-based Employee Onboarding and Offboarding Workflow Tracker built with Python Flask and SQLite. HR staff can add employees, start an onboarding or offboarding process, and generate a checklist automatically from reusable task templates. Each task is assigned to a department, given a due date and updated as Pending, In Progress or Completed. The system shows each employee's completion percentage, highlights overdue tasks on a dashboard and keeps a history of status changes. Role-based login separates HR administrators from department staff, who see only their own tasks. HR can search records and export reports. It aims to reduce missed steps and save HR time.

3. Objectives

The main objectives of this project are:
• To implement secure login for HR Administrators and Department Staff using hashed passwords.
• To provide employee record management features, including adding, viewing, editing, and searching records.
• To create reusable task templates and automatically generate checklists with due dates.
• To assign tasks to relevant departments and track their status.
• To develop a dashboard displaying employee process completion percentages and overdue tasks.
• To maintain a history of task status changes, including the user and date/time.
• To allow HR staff to export process reports in CSV format.

• To test the application using at least 10 documented test cases and sample data covering at least three departments, and collect feedback from at least five evaluators.

4. Scope

The project will focus on managing employee onboarding and offboarding activities through a centralized web application.
The system will include:
• Employee record management.
• Onboarding and offboarding process management.
• Reusable task templates and automatic checklist generation.
• Department-wise task assignment.
• Task status and due-date tracking.
• Role-based access for HR Administrators and Department Staff.
• A dashboard for completion progress and overdue tasks.
• Task status history and CSV report export.
The project will be limited to a student-level web application using Flask and SQLite. It will not include artificial intelligence, paid external services, or integrations with external HR platforms.

5. Expected Outcomes

The expected outcome is a functional web application that helps organize employee onboarding and offboarding procedures.
The system is expected to:
• Reduce missed tasks through standardized checklists.
• Improve visibility of task progress and overdue activities.
• Make departmental responsibilities clearer.
• Help HR staff maintain organized employee and process records.
• Improve accountability through task status history.
• Provide convenient reporting through CSV exports.
These outcomes will be evaluated through functional testing and feedback from evaluators.

6. Technologies

The following technologies and tools will be used:
• **Programming Language:** Python 3
• **Web Framework:** Flask
• **Templating Engine:** Jinja2
• **Database:** SQLite using Python's built-in sqlite3 module
• **Frontend:** HTML5 and CSS3
• **Responsive Interface:** Bootstrap 5
• **Security:** Werkzeug for password hashing and session-related functionality
• **Version Control:** Git and GitHub
• **Development Environment:** Visual Studio Code
• **Database Tool:** DB Browser for SQLite
• **Testing Browser:** Google Chrome or Microsoft Edge
• **Architecture:** The application will follow the Model-View-Controller (MVC) pattern, with Flask route functions

handling application requests, Jinja2 templates displaying the interface, and model/database logic managing SQLite data.

• **Hardware Requirements:** A standard laptop or desktop computer capable of running Python, Flask, SQLite, and a modern web browser.